Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 05: Dispersion 1

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A simple model of harmonically bound charges

$\vec{v} \times \vec{B} \ll \vec{E} \Rightarrow |\vec{v}| \ll c$

- Neglect magnetic fields and Lorentz Force, because the speed is much smaller than the speed of light.

$$m\vec{a} = -e\vec{E}'(x,t) + \vec{F}_b + \vec{F}_{damp}$$

$$\vec{F}_b = -m\omega_0^2 \vec{x}, \quad \vec{F}_{damp} = -\gamma m \vec{v}$$

$$m(\ddot{\vec{x}} + \gamma \dot{\vec{x}} + \omega_0^2 \vec{x}) = -e\vec{E}'(x,t)$$

$$\vec{x} \propto e^{-i\omega t} \qquad \frac{d\vec{x}}{dt} = -i\omega \vec{x}$$

$$\Rightarrow \quad m(-\omega^2 \vec{x} - i\omega\gamma \vec{x} + \omega_0^2 \vec{x}) = -e\vec{E}'$$

A simple model of harmonically bound charges

$\vec{v} \times \vec{B} \ll \vec{E} \Rightarrow |\vec{v}| \ll c$

- Neglect magnetic fields and Lorentz Force, because the speed is much smaller than the speed of light.

$$\vec{x} = -\frac{e}{m(\omega_0^2 - \omega^2 - i\omega\gamma)} \vec{E}'$$

For each charge, $\vec{p}_{mol} = -e\vec{x}$

$$= \frac{e^2}{m(\omega_0^2 - \omega^2 - i\omega\gamma)} \vec{E}'$$

$$\vec{p} \propto \vec{E}$$

Molecular polarizability (γ_{mol})

- The ratio of the average molecular dipole moment to ε_0 times the applied field E' (microscopic) at the molecule.

$$\gamma_{mol} = \frac{\sum_j \vec{p}_{mol,j}}{\varepsilon_0 \vec{E}'} = \sum_j \frac{e^2}{\varepsilon_0 m(\omega_{0j}^2 - \omega^2 - i\omega\Gamma_j)}$$

j means different electrons.

- Polarization vector?

$$\vec{P} = N \sum_j \vec{p}_{mol,j}$$

N is molecular number density.

$$= N \varepsilon_0 \gamma_{mol} \vec{E}'$$

Dielectric constant, Susceptibility, and polarizability σ_{mol}

- Low density

$N \ll 1$. Macroscopic $\vec{E}$

Microscopic $\vec{E}'$

$\vec{E} \approx \vec{E}'$ No impact from surrounding molecules.

$\vec{P} = \varepsilon_0 \chi_e \vec{E}$ $\quad \vec{P} = N \sum_j \vec{p}_{mol,j} = N \varepsilon_0 \sigma_{mol} \vec{E}$

$\chi_e = N \sigma_{mol}$

Dielectric constant, Susceptibility, and polarizability σ_{mol}

- Low density

$N \ll 1$. $\chi_e = N \sigma_{mol}$

$$\frac{\varepsilon}{\varepsilon_0} = 1 + \chi_e = 1 + N\sigma_{mol}$$

$$= 1 + \frac{Ne^2}{\varepsilon_0 m(\omega_0^2 - \omega^2 - i\omega\Gamma)}$$

Assume each molecule has one electron.

$$= 1 + \frac{Ne^2}{\varepsilon_0 m} \sum_j (\omega_{0j}^2 - \omega^2 - i\omega\Gamma_j)^{-1}$$

Dielectric constant, Susceptibility, and polarizability

- High density $N >> 1$

$$\vec{p}' \neq \vec{p} \qquad \vec{p}' = \vec{p} - \vec{p}_{in}$$

$$\chi_e = \frac{N\alpha_{mol}}{1 - \frac{N\alpha_{mol}}{3}}$$

Resonant frequency

$$\frac{1}{\omega_{0j}^2 - \omega^2 - i\omega\gamma_j} = \frac{\omega_{0j}^2 - \omega^2 + i\omega\gamma_j}{(\omega_{0j}^2 - \omega^2)^2 + \omega^2\gamma_j^2}$$

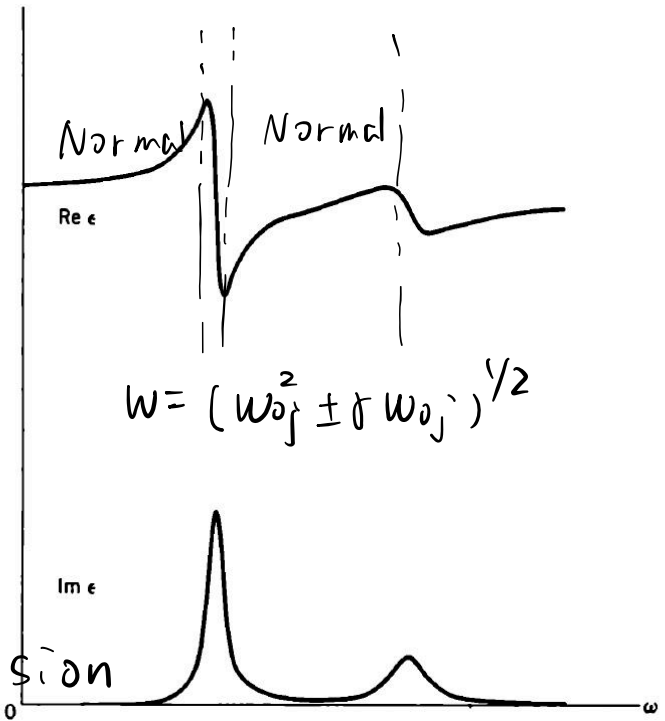
$$\frac{\epsilon}{\epsilon_0} = 1 + \frac{Ne^2}{\epsilon_0 m} \sum_j (\omega_{0j}^2 - \omega^2 - i\omega\gamma_j)^{-1}$$

When $\omega \neq \omega_{0j}$, $-i\omega\gamma_j$ can be neglected.

$$\frac{\epsilon}{\epsilon_0} = 1 + \frac{Ne^2}{\epsilon_0 m} \sum_j (\omega_{0j}^2 - \omega^2)^{-1}$$

When $\omega \uparrow$, $\epsilon \uparrow$, normal dispersion

When $\omega \uparrow$, $\epsilon \downarrow$, anomalous dispersion



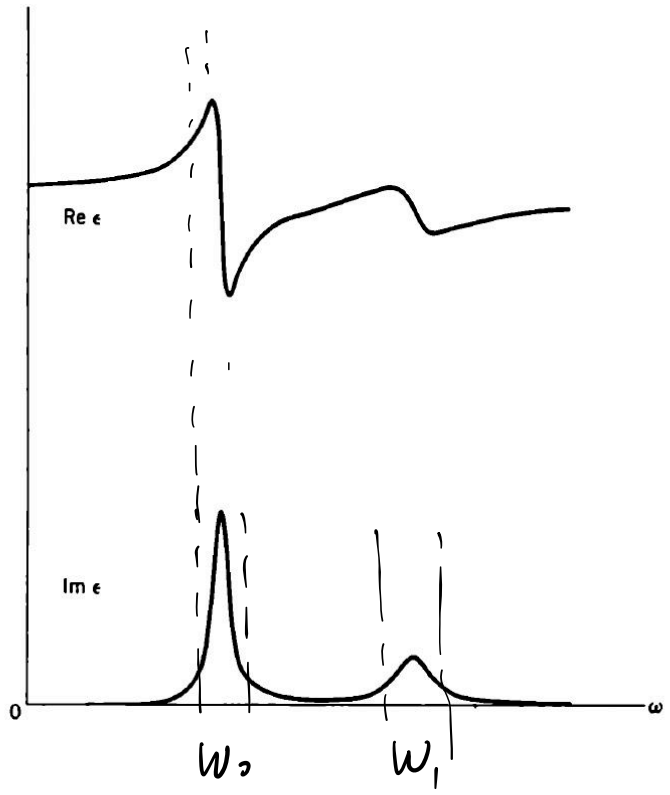
$$\omega = (\omega_{0j}^2 \pm \gamma\omega_{0j})^{1/2}$$

Resonant frequency

$$\frac{\epsilon}{\epsilon_0} = 1 + \frac{Ne^2}{\epsilon_0 m} \sum_j (\omega_{0j}^2 - \omega^2 - i\omega\sigma_j)^{-1}$$

When $\omega \sim \omega_{0j}$

Resonant absorption.



Low-frequency behavior ($\omega \sim 0$)

- In insulators ($\omega_j > 0$)

$$\frac{\epsilon}{\epsilon_0} = 1 + \frac{Ne^2}{\epsilon_0 m} \sum_j \left(\omega_{0j}^2 - \omega^2 - i\omega\gamma_j\right)^{-1}$$

Assume $\gamma_j \ll \omega_{0j}$

$$= 1 + \frac{Ne^2}{\epsilon_0 m} \sum_j \frac{1}{\omega_{0j}^2}$$

Low-frequency behavior ($\omega \sim 0$)

- In conductors ($\omega_{00} = 0$)

One free electron

$$\frac{\varepsilon}{\varepsilon_0} = 1 + \underbrace{\frac{Ne^2}{\varepsilon_0 m} \sum_{j>0} \frac{1}{\omega_{0j}^2}}_{\text{binding electrons}} + \underbrace{\frac{Ne^2}{\varepsilon_0 m} (-\omega^2 - i\gamma_0 \omega)^{-1}}_{\text{free electron}}$$

$$\varepsilon = \varepsilon_b + \frac{i Ne^2}{m\omega(\gamma_0 - i\omega)}$$

Low-frequency behavior ($\omega \sim 0$)

$$\vec{E} \propto e^{-i\omega t}.$$

- Ohm's Law

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} = \varepsilon \frac{\partial \vec{E}}{\partial t} = -i\omega \varepsilon \vec{E}$$

$$= -i\omega \left(\varepsilon_b + \frac{i Ne^2}{m\omega(\tau_0 - i\omega)} \right) \vec{E}$$

$$= -i\omega \varepsilon_b \vec{E} + \frac{Ne^2}{m(\tau_0 - i\omega)} \vec{E} \qquad \vec{D}_b = \varepsilon_b \vec{E}$$

$$= \frac{\partial \vec{D}_b}{\partial t} + \sigma \vec{E} \qquad \sigma = \frac{Ne^2}{m(\tau_0 - i\omega)} \qquad \varepsilon = \varepsilon_b + i\frac{\sigma}{\omega}$$

$$+ \vec{J}$$

Low-frequency behavior ($\omega \sim 0$)

- Conductivity

$$\omega \to 0, \quad \sigma = \frac{Ne^2}{m\gamma_0}$$

Copper. $N = 8 \times 10^{28}\ \#/m^3$ $\sigma = 5.9 \times 10^7\ (\Omega \cdot m)^{-1}$

$\gamma_0 = 4 \times 10^{13}$ Hz

When $\omega < \gamma_0$, σ is almost real.